



Molecular-based characterization and bioengineering of *Sorghum bicolor* to enhance iron deficiency tolerance in iron-limiting calcareous soils

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Received: 1 October 2023 / Accepted: 16 September 2024 / Published online: 24 October 2024
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Abstract

Plant biomass can significantly contribute to alternative energy sources. *Sorghum bicolor* is a promising plant for producing energy, but is susceptible to iron deficiency, which inhibits its cultivation in iron-limiting calcareous soils. The molecular basis for the susceptibility of sorghum to iron deficiency remains unclear. Here, we explored the sorghum genome to identify genes involved in iron uptake and translocation. Iron deficiency-responsive gene expression was comparable to that in other graminaceous plants. A nicotianamine synthase gene, *SbNAS1*, was induced in response to iron deficiency, and *SbNAS1* showed enzyme activity. Sorghum secreted 2'-deoxymugineic acid and other phytosiderophores under iron deficiency, but their levels were relatively low. Intercropping of sorghum with barley or rice rescued iron deficiency symptoms of sorghum. To produce bioengineered sorghum with enhanced tolerance to iron deficiency, we introduced four cassettes into sorghum: *35S* promoter-*OsIRO2* for activation of iron acquisition-related gene expression, *SbIRT1* promoter-*Refre1/372* for enhanced ferric-chelate reductase activity, and barley *IDS3*, and *HvNAS1* genomic fragments for enhanced production of phytosiderophores and nicotianamine. The resultant single sorghum line exhibited enhanced secretion of phytosiderophores, increased ferric-chelate reductase activity, and improved iron uptake and leaf greenness compared with non-transformants under iron-limiting conditions. Similar traits were also conferred to rice by introducing the four cassettes. Moreover, these rice lines showed similar or better tolerance in calcareous soils and increased grain iron accumulation compared with previous rice lines carrying two or three comparable cassettes. These results provide a molecular basis for the bioengineering of sorghum tolerant of low iron availability in calcareous soils.

Key message

Molecular and physiological responses of sorghum to iron deficiency were comparable to those in rice. Bioengineering of sorghum and rice for iron uptake and translocation enhanced iron deficiency tolerance.

Keywords Bioengineering · Calcareous soil · Energy plant · Iron deficiency tolerance · Rice · Sorghum bicolor

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Introduction

Global warming due to an increasing concentration of CO₂ in the atmosphere is a serious problem [Intergovernmental Panel on Climate Change (IPCC), 2023]. The Intergovernmental Panel on Climate Change (IPCC) noted in the sixth assessment report that the main cause of this increase is the burning of fossil fuels. The development and promulgation of more environmentally friendly and sustainable energy systems are required to reduce the dependence on fossil fuels. As an alternative energy source, plant biomass is attracting attention.

Sorghum [*Sorghum bicolor* (L.) Moench] is the fifth most cultivated cereal crop in the world after wheat, maize, rice, and barley in order of production area (FAOSTAT 2012). Sorghum is cultivated on about 40 million hectares worldwide, mainly in arid, and semiarid regions of Africa, Asia, and the Americas. Sorghum is cultivated for human food, animal feed, fiber, and syrup, among other uses. In addition, sorghum has recently been considered as a promising biomass plant for biofuel production (van der Weijde et al. 2013). Sorghum is a C4 plant, so it can grow effectively at high temperatures and in arid conditions compared with C3 plants (Hatch & Slack 1966). In particular, the high drought tolerance of sorghum enables it to be cultivated in marginal regions where other cereals cannot be cultivated. Sorghum grows vigorously and some cultivars reach more than 4 m within 4 months (Yu et al. 2008). Furthermore, cultivars commonly called “sweet sorghum” accumulate sugar in the culm like sugarcane. The monetary and energetic costs of fermentation from the juice are lower than those from lignocellulosic carbon sources. These advantages have attracted molecular breeding of sorghum for biomass production.

One of the problems limiting the cultivation area of plants is susceptibility to iron (Fe) deficiency (Clark 1982). Fe is essential to all living organisms, including plants, for many biological reactions as a cofactor of enzymes and electron carrier. Although Fe is abundant in the soil, Fe that is available to plants is scarce in high-pH conditions where Fe is nearly water-insoluble. Thus, plants often lapse into Fe deficiency in alkaline conditions such as calcareous soils. The most obvious symptom of Fe deficiency is chlorosis in new leaves, which results in insufficient photosynthesis and consequently leads to growth defects and yield loss. Calcareous soil is broadly distributed in arid and semiarid mid-latitude regions, accounting for approximately 30% of cultivated soils worldwide, where impaired production of crops is a major agricultural problem (Tester and Langridge 2010).

To acquire poorly soluble Fe, plants have developed unique Fe acquisition systems, which are activated under

Fe deficiency. Nongraminaceous plants utilize strategy I, or reduction mechanism (Marschner et al. 1986; Römheld and Marschner 1986). In this system, ferric-chelate reductase (FRO) on the root surface reduces Fe(III)-chelate to Fe(II), which is then absorbed through the Fe²⁺ transporter IRT (Eide et al. 1996; Robinson et al. 1999). Graminaceous plants have another Fe acquisition mechanism called strategy II, or the chelation mechanism (Marschner et al. 1986; Römheld and Marschner 1986). This system relies on the biosynthesis and secretion of potent Fe(III) chelators called phytosiderophores (PS) or mugineic acid family PSs (MAs) from root to rhizosphere, where they mobilize Fe from soil as Fe(III)-MAs (Takagi 1976). MAs are biosynthesized from methionine (Mori and Nishizawa 1987) through the condensation of three *S*-adenosyl-L-methionine molecules to produce nicotianamine (NA) catalyzed by NA synthase (NAS) (Higuchi et al. 1999), followed by sequential reactions of NA aminotransferase (NAAT; Takahashi et al. 1999) and DMA synthase (DMAS; Bashir et al. 2006). In rice, 2'-deoxymugineic acid (DMA) is the major MA and is released to the rhizosphere through OsTOM1 transporter (Nozoye et al. 2011). In addition to DMA, barley secretes other MAs such as mugineic acid (MA), 3-epihydroxy-2'-deoxy mugineic acid (epiHDMA), and 3-epihydroxy-mugineic acid (epiHMA) (Kawai et al. 1988; Ma et al. 1999). These MAs are deduced to be synthesized from DMA by IDS2 and IDS3 dioxygenases (Nakanishi et al. 2000; Kobayashi et al. 2001). Fe(III)-MAs are absorbed into root cells through Yellow Stripe (YS1)/YS1-like (YSL) transporters (Curie et al. 2001, 2009; Inoue et al. 2009). Recently, an analog of DMA, designated proline-DMA (PDMA), was shown to solubilize Fe and remedy Fe deficiency symptoms in plants such as rice grown in calcareous soils (Suzuki et al. 2021; Ueno et al. 2021; Wang et al. 2023). Administration of unchelated PDMA and Fe(III)-chelated PDMA into calcareous soils similarly improves rice Fe concentration and chlorosis symptoms (Suzuki et al. 2021), indicating that Fe deficiency due to its low solubility is a major limiting factor in plant growth in calcareous soils.

After absorption from the rhizosphere, internal translocation of Fe is mediated by chelators and transporters. NA is not only an intermediate for MA production, but is also a potent metal chelator for the internal translocation of Fe and other divalent metals, and plays a crucial role in all plants for internal metal translocation (Scholz et al. 1992; Inoue et al. 2003; Curie et al. 2009). As an efflux transporter of NA, OsENA1 was identified in rice (Nozoye et al. 2011), which is thought to be involved in Fe translocation. MAs also participate in internal Fe translocation in graminaceous plants (Kobayashi et al. 2019a). Cellular uptake of Fe(III)-MAs

and Fe(II)-NA for Fe translocation is mediated by YS1/YSL transporters (Curie et al. 2009; Kobayashi et al. 2019a).

The expression of key genes for Fe uptake and translocation is basically regulated at the transcriptional level in plants, although protein-level regulation is also crucial (Kobayashi and Nishizawa 2012; Kobayashi et al. 2019a). In graminaceous plants, the expression of genes involved in the biosynthesis and secretion of MAs, as well as those for Fe(III)-MA uptake, is strongly induced by Fe deficiency. Several key regulators of such Fe deficiency responses have been identified in rice, such as transcription factors IDEF1, IDEF2, OsIRO2, OsIRO3, OsbHLH058, OsbHLH059, OsbHLH060, OsFIT, and OsbZIP83, ubiquitin ligase OsHRZs, an aconitase OsACO1, and Fe deficiency-inducible peptides OsIMAs (Kobayashi et al. 2014, 2019b, 2021, 2022b; Kobayashi 2019; Senoura et al. 2020). Among these, OsIRO3, and OsHRZs negatively regulate subsets of Fe deficiency-inducible genes for Fe uptake and translocation, whereas others positively regulate subsets of these genes.

There is a large genetic variation in plants regarding tolerance to low Fe availability. In graminaceous species, tolerance to Fe deficiency is strongly correlated to the amount of MAs biosynthesized and secreted, for which barley, and wheat outperform other common crops such as rice, maize, and sorghum (Römheld and Marschner 1986; Marschner 1995; Mori et al. 1991). To date, the susceptibility of rice plants to Fe deficiency has been successfully improved by bioengineering various genes involved in Fe uptake, translocation, and their regulation. Effective methods include: (i) introduction of genomic fragments of biosynthetic genes for MAs, such as *HvNAAT-A*, *-B*, *HvNAS1*, and *IDS3*, from barley, which produces large amounts and many kinds of MAs (Takahashi et al. 2001; Suzuki et al. 2008); (ii) enhancement of gene expression of transcription factors positively regulating Fe deficiency responses in rice, such as IDEF1 and OsIRO2 (Kobayashi et al. 2007; Ogo et al. 2007, 2011); (iii) knocking down of negative regulators of Fe deficiency responses in rice, OsHRZs (Kobayashi et al., 2013); and (iv) introduction of a ferric-chelate reductase gene, *Refre1/372*, which is a mutationally reconstructed artificial gene originating from the yeast ferric-chelate reductase gene *FRE1* (Dancis et al. 1990) to optimize codon usage for plants and to confer enhanced reductase activity even under alkaline conditions (Oki et al. 2004). The introduction of *Refre1/372* downstream of the rice Fe²⁺ transporter gene *OsIRT1* promoter was highly effective at conferring tolerance to low Fe availability in calcareous soils (Ishimaru et al. 2007). In addition, the combined introduction of the above genes, such as (ii) + (iv), (i) + (ii) + (iv), or (iii) + (iv), further improved tolerance or robustness under various growth conditions in

calcareous soils (Masuda et al. 2017, 2019; Kobayashi et al. 2022a). Furthermore, the bioengineering of genes involved in Fe uptake, translocation, and storage also enhances Fe accumulation in the edible parts of crops. For instance, Fe biofortification in rice seeds was successfully achieved by seed-specific expression of the Fe storage protein ferritin, or enhancement of Fe flow into seeds by the overexpression of *NAS* genes, phloem-specific expression of the Fe(II)-NA transporter gene *OsYSL2*, knocking down of the negative regulator *OsHRZs*, or their combinations (Ishimaru et al. 2010; Masuda et al. 2012; Kawakami and Bhullar 2021; Kobayashi et al., 2013, 2022a). These results indicate the potency of knowledge-based bioengineering for improving growth traits.

In the present report, we aimed to develop sorghum with increased Fe deficiency tolerance to extend the area where sorghum can be cultivated, including calcareous soils not suitable for agriculture. This is worthwhile because the cultivation of energy plants should not compete with crop production. As the mechanism of Fe acquisition in sorghum has not been clarified at the genetic or molecular level, we first searched for potential counterparts of rice genes known to be responsible for Fe acquisition in sorghum, based on the available whole-genome sequence of sorghum cv. BTx623 (Paterson et al. 2009), and confirmed their expressional response to Fe deficiency. Physiological responses of sorghum to Fe-deficient conditions were then analyzed for the secretion of MAs and ferric-chelate reduction. From these results, gene cassettes for the improvement of Fe deficiency tolerance were constructed and introduced into sorghum. One transgenic sorghum line, as well as multiple rice lines introduced with the same cassettes, showed enhanced tolerance to low Fe availability in calcareous soils, along with the improved secretion of MAs. Our results provide the fundamental basis for future bioengineering of crop plants for enhanced biomass production in calcareous soils.

Materials and methods

Hydroponic culture of sorghum and rice

Seeds of *Sorghum bicolor* cv. BTx623 were obtained from National Institute of Agrobiological Sciences of Japan. For gene expression analysis and ferric-chelate reductase activity assay, the seeds were germinated on moist paper for 6 days in a growth chamber at 28 °C (16 h light/8 h dark cycle), and hydroponically cultivated in a glass house (natural light 14 h at 30 °C/dark 10 h at 25 °C cycle) with continuous air bubbling. For hydroponic cultivation, 4 plants were planted in a box containing 5 L of Kasugai's solution (pH 5.5) with the following concentrations: 0.70 mM K₂SO₄, 0.10 mM

KCl, 0.10 mM KH_2PO_4 , 2.0 mM $\text{Ca}(\text{NO}_3)_2$, 0.50 mM MgSO_4 , 10 μM H_3BO_3 , 0.50 μM MnSO_4 , 0.20 μM CuSO_4 , 0.50 μM ZnSO_4 , 0.050 μM Na_2MoO_4 , and 0.10 mM Fe(III)-EDTA. The pH was adjusted to 5.5 with HCl every day. Fe was omitted for Fe-deficient treatments. For measurement of MA secretion, sorghum seeds were germinated in 1/2 Murashige and Skoog medium containing 2.0% Gellan Gum (Fujifilm-Wako, Tokyo, Japan) for 12 days and transferred to hydroponic culture as described above. Fe deficiency was initiated 40 days after germination by washing the roots and containers with deionized water and transferring the plants into a new hydroponic solution without Fe and imposed by further cultivation for 7 days. For ferric-chelate reductase assay, 14-day-old seedlings were treated with or without Fe for 4 days as above.

Regenerated transgenic sorghum plants (defined as T_0 generation) were grown to obtain seeds (T_1 generation). T_1 seeds were surface-sterilized in 70% ethanol for 2 min and 2.0% sodium hypochlorite for 25 min, and sown on MS medium with 30 $\mu\text{g/L}$ hygromycin B to select transformants. After cultivation at 28 °C for 10–21 days, hydroponic culture was carried out as above under Fe-free solution for 18 days for the measurement of MAs secretion and ferric-chelate reductase activity. T_1 transformants after hydroponic culture were also cultivated in calcareous soils as below to obtain T_2 seeds (the next generation), which were analyzed as above.

Transgenic and wild-type rice (*Oryza sativa* L. cv. Tsukinohikari) seeds were germinated and cultured hydroponically as described by Kobayashi et al. (2019b). For the measurement of MA secretion, seedlings (11 days old for Fig. 3b and 17 days old for Fig. S10) were grown hydroponically under control solution for 7 days for Fig. 3b and 21 days for Fig. S10, and then under Fe-free solution for 7 days. For ferric-chelate reductase assay, seedlings (17 days old) were grown hydroponically under control solution for 14 days, and then under Fe-free solution for 10 days. For calcareous soil tests, seedlings (16–17 days old) were precultured hydroponically for 5–7 days.

Gene expression analysis

Total RNA was prepared from sorghum roots and shoots using RNeasy Plant Mini Kit (Qiagen, Venlo, Netherlands), in accordance with the manufacturer's instructions. cDNA was synthesized using ReverTraAce qPCR RT Master Mix with gDNA Remover (Toyobo, Osaka, Japan). Quantitative real-time PCR (RT-qPCR) was performed using StepOnePlus® Real-Time PCR System (Applied Biosystems, Massachusetts, USA) with SYBR Green I and the ExTaq® Real-Time-PCR version (TaKaRa Bio Inc., Shiga, Japan). A *tubulin* gene (Sb09g004420), whose expression level was comparable irrespective of Fe

conditions, was used for normalization. Sequences of the primers are listed in Table S1.

NAS enzyme assay

The cDNA fragment containing full-length open reading frame (ORF) of *SbNAS1* was amplified by PCR using primers listed in Table S1. The fragment between the *EcoRI* and *HindIII* sites was inserted into the same sites of pMAL-c2 (New England Biolabs, Massachusetts, USA) to fuse with maltose-binding protein (MBP) at the N-terminus of *SbNAS1*. The construct was introduced into *Escherichia coli* strain BL21(DE3)pLysS, and recombinant MBP-*SbNAS1* protein was obtained using the MBP purification system, in accordance with the manufacturer's instructions. MBP alone (empty pMAL-c2) and MBP-HvNAS1 (Higuchi et al. 1999) were prepared by the same procedure. Ten micrograms of each purified protein was used for an enzyme assay by thin-layer chromatography, as described previously (Higuchi et al. 1999; Mizuno et al. 2003).

Analysis of MA secretion

Root washings were collected from sorghum, rice, or *Brachypodium distachyon* in purified water for 6 h after the initial illumination. MAs in the root washings were determined by high-performance liquid chromatography, as described previously (Mori et al. 1987; Mori and Nishizawa 1987; Suzuki et al. 2006; Nozoye et al. 2011, 2013, 2014). Sorghum root exudates were collected for 6 h from the start of illumination. Briefly, the roots of 2–3 plants were transferred to 600 mL of distilled water at the beginning of the illumination. Distilled water was replaced every 2 h, and a total of 1.8 L of distilled water was used for the analysis of MAs. The experiment was carried out with five replicates. For *Brachypodium distachyon* cv. BD21, seeds were germinated on the wetted paper at 4 °C overnight, and moved to room temperature and grown hydroponically as described above at 25 °C (14 h light/10 h dark cycle). Fe deficiency was initiated at 20 days after germination. Three and 11 days after –Fe deficiency treatment, root exudates were collected, and combined to use for analysis of MAs. Rice root exudates were collected for 6 h from the start of illumination.

Ferric-chelate reductase activity assay

Ferric-chelate reductase activity was measured as described by Oki et al. (2004) and Ishimaru et al. (2007). Whole roots of sorghum or rice were submerged in 30–40 mL of assay solution [5 mM HEPES (pH 5.5), 0.2 mM bathopenanthrolinedisulfonic acid disodium salt, 0.2 mM CaSO_4 , 0.1 mM Fe(III)-EDTA] in a growth chamber at

28 °C for 1–5 h in the dark. Then, A_{535} of the assay solution was measured. A standard curve was developed with known concentrations of FeSO_4 .

Intercropping analysis

Sorghum plants were intercropped with rice and barley as described by Xiong et al. (2013) and Nozoye et al. (2017). For rice, we used non-transformants (NT) and a transgenic line 36 carrying barley genomic fragment of *IDS3* (Kobayashi et al. 2001). Sorghum and barley (*Hordeum vulgare* cv. Ehimehadaka) seeds were germinated on wet paper for 18 days under a 14 h light and 10 h dark cycle at 30 °C for sorghum and 25 °C for barley. Rice seeds of *IDS3* line 36 and NT were germinated on MS plates with and without 50 mg/L hygromycin B, respectively, for a week under a 14 h light and 10 h dark cycle at 30 °C. These sorghum, barley, and rice plants were transferred separately to Kasugai's nutrient solution described above with continuous air bubbling in a greenhouse at 30 °C under a 14-h/10-h light/dark photoperiod under supplemented natural light. Twenty-one days after germination, intercropping sorghum, and barley or rice was initiated with nutrient solution supplying $\text{Fe}(\text{OH})_3$ as the sole Fe source. Soil Plant Analysis Development (SPAD) values of the leaves were measured by SPAD-502Plus (KONICA MINOLTA, Tokyo, Japan).

Production of transgenic sorghum and rice plants

Construction of the vector used for sorghum and rice transformation, designated as pBGRZ-*OsIRO2-Refre1-IDS3-HvNAS1*, is described in Fig. S1. Sorghum (cv. Tx430) transformation was carried out in accordance with a previously described method (Sato-Izawa et al. 2018). Pretreatment of immature embryos was performed by centrifugation for 10 min and heating (43 °C) for 15 min prior to *Agrobacterium* infection. One line of transformant carrying pBGRZ-*OsIRO2-Refre1-IDS3-HvNAS1* (line N22) and one line of vector control carrying pBGRZ1 (line VC) were obtained (T_0 generation). Using these lines, T_1 and T_2 generations were obtained as above.

Rice (*Oryza sativa* L. cv. Tsukinohikari) transformation was carried out as described previously (Hiei et al. 1994; Kobayashi et al. 2001). T_1 transformants were cultivated in calcareous soils as below to obtain T_2 seeds.

Southern hybridization analysis

Genomic DNA was isolated from fresh leaves of a N22 plant (T_0 generation) using the cetyltrimethylammonium bromide (CTAB) method described by Murray and Thompson (1980). Twelve micrograms of genomic DNA was digested

by *Hind*III, separated by electrophoresis in a 0.8% agarose gel, and transferred to a nylon membrane, Hybond-XL, using an alkaline method. A 650-bp probe of the hygromycin resistance gene (*HPT*) was prepared from pIG121Hm vector (Hiei et al. 1994) using a PCR DIG Probe synthesis kit and the primers listed in Table S1. Hybridization, washing, and detection procedures were carried out according to the instruction manual of the DIG labeling and detection system (Roche Diagnostics). DIG signal was visualized using LAS4000mini (Fuji Film).

Genomic PCR analysis

Sorghum genomic DNA was extracted from each 4-day-old seedling of T_2 transgenic N22 or VC line germinated on moist paper, using Template Prepper for DNA (NipponGene). The transgenes were amplified by genomic PCR using KOD One PCR Master Mix (Toyobo, Osaka, Japan) and the primers listed in Table S1. The sequence of the amplified fragments were confirmed by sequencing around the both primer positions.

Rice genomic DNA was extracted from 1 cm leaf segments of T_0 transgenic lines and NT by immersing into 100 μL of extraction buffer [100 mM tris(hydroxymethyl) aminomethane (pH 9.5), 1 M KCl, 10 mM EDTA] and incubating at 95 °C for 10 min. The transgenes and *OsActin* gene as an internal control were amplified by genomic PCR using KOD FX Neo (Toyobo, Osaka, Japan) and the primers listed in Table S1 following the manufacturer's instructions (Toyobo, Osaka, Japan).

Calcareous soil culture of sorghum and rice

Cultivation in calcareous soil pots was carried out in a glass house under a cycle of natural light for 14 h at 30 °C and dark for 10 h at 25 °C, as described by Kobayashi et al. (2013) with the following modifications. For sorghum culture, T_1 and T_2 seeds were germinated for 10–14 days, and grown hydroponically for 10 days as described above. These seedlings were grown in 1 L pots for T_1 analysis, or 2 L pots for T_2 analysis, filled with a calcareous soil from Takaoka City, Toyama, Japan (pH 9.0–9.6; sandy calcareous soil containing fossil shell; Nihonkai Hiryou, Takaoka, Japan; Morikawa et al. 2004) with 3 g each of controlled-release fertilizers EcoLongTotal 70 and LongTotal 140 (JCAM AGRI. CO., LTD., Tokyo, Japan; Masuda et al. 2019), and were watered every 2–3 days up to 5 cm from the bottom.

For rice culture, we used pBGRZ-*OsIRO2-Refre1-IDS3-HvNAS1* lines (designated “Q” lines for quadruple; T_1 , T_2 , and T_3 seeds), as well as previously characterized and selected lines of double cassettes (designated “line D” for double; T_3 seeds of line 22 of 35S

promoter-*OsIRO2* + *OsIRT1* promoter-*Refre1/372*; Masuda et al. 2017), and triple cassettes (designated “line T” for triple; T₃ seeds of line 4 of 35S promoter-*OsIRO2* + *OsIRT1* promoter-*Refre1/372* + *IDS3* genomic fragment; Masuda et al. 2019), and NT. These seeds were germinated and pre-cultured hydroponically as described above, until the plant height reached about 30 cm for T₁ and T₂ analyses shown in Fig. S8 and S9, or 20–25 cm for T₃ analysis in Fig. 8. These seedlings were grown in 1 L pots filled with a calcareous soil as above, with 3 g each of EcoLongTotal 70 and LongTotal 140 controlled-release fertilizers for T₁ and T₂ analyses, or 0.85 g each for T₃ analysis, as 1 seedling/pot for T₁ and T₃ analyses, or 3 seedlings/pot for T₂ analysis. Water was supplied every 2–3 days up to 5 cm from the bottom for T₁ and T₂ analyses, or up to half of the pot height for T₃ analysis.

Metal concentration analysis

For sorghum plants, roots and the youngest leaves were collected from plants grown hydroponically under Fe sufficiency or deficiency for 10 days. The samples were dried at 50 °C for 4 days, and aliquots of approximately 50 µg for roots and 100 µg for leaves were cut into small pieces and digested with 3.0 mL of 13.4 M nitric acid for 20 min at 200 °C using a MarsXpress oven (CEM, Matthews, NC, USA). For rice seeds, 6–10 brown seeds for each replicate were dehusked and dried for 2 d at 70 °C, weighed, and wet-washed with 1.0–1.5 mL of 13.4 M nitric acid and 1.0–1.5 mL of 8.8 M hydrogen peroxide for 20 min at 220 °C using a MarsXpress oven. The digested solutions were diluted and filtered through a 0.20 µm filter (Sartorius, Gottingen, Germany). Fe, zinc (Zn), manganese (Mn), and copper (Cu) concentrations were measured using inductively coupled plasma optical emission spectrometry (ICPS-8100, Shimadzu, Kyoto, Japan, for sorghum; 5800 ICP-OES, Agilent, Tokyo, Japan, for rice).

Results

Expression profile of putative Fe acquisition-related genes in response to Fe status

We searched for sorghum genes closely related to rice genes, which are responsible for Fe uptake and translocation. Amino acid sequences of 15 rice genes were applied to BLASTP programs and 17 candidates of sorghum counterparts were selected according to the primary structure relationships (Materials and Methods S1, Results S1, Table S2, Fig. S2).

Using RT-qPCR analysis, we investigated the expression levels of these genes in sorghum roots and leaves cultured hydroponically under Fe sufficiency and deficiency (Fig. 1). Among the putative genes for biosynthesis of MAs, *SbNAS1*

was strongly induced (> 45-fold) in roots by Fe deficiency (Fig. 1a), but the expression and induction levels in shoots were lower (Fig. 1b). In contrast, *SbNAS3a* and *SbNAS3b* were expressed at apparently higher levels in shoots than in roots, and were significantly upregulated in shoots but downregulated in roots by Fe deficiency. Similar to *SbNAS1*, genes possibly involved in DMA synthesis, *SbNAAT1a*, *SbNAAT1b*, and *SbDMAS1*, were upregulated strongly (18-, 6.0-, and 11-fold, respectively) in Fe-deficient roots and moderately in Fe-deficient shoots (2.5-, 1.6-, and 1.3-fold, respectively). In addition, *SbTOM1*, *SbYSL15a*, *SbYSL15b*, and *SbYSL15c*, possibly involved in the secretion of MAs and uptake of Fe(III)-MAs in roots, were upregulated in Fe-deficient roots (84-, 4.6-, 2.9-, and 5.6-fold, respectively), but the expression of these genes in shoots was not detected regardless of Fe status. In contrast, *SbYSL16* was expressed more highly in shoots than in roots, and was moderately upregulated by Fe deficiency in shoots but only slightly in roots. *SbENAI* was strongly upregulated in both roots and shoots by Fe deficiency (9.6- and 116-fold, respectively), suggesting that NA is required in Fe-deficient conditions in both roots and shoots. *SbFRO1* was expressed more highly in shoots than in roots, consistent with a previous report (Mikami et al. 2011) and similar to previously reported findings for *OsFRO1* (Ishimaru et al. 2006). Under Fe deficiency, *SbFRO1* was upregulated 1.7-fold in shoots but was unchanged in roots, suggesting that *SbFRO1* makes little, if any, contribution to ferric-chelate reduction in roots. *SbIRT1*, *SbIRT2a*, and *SbIRT2b* were strongly upregulated by Fe deficiency in both roots and shoots. *SbIRO2* was also upregulated by Fe deficiency in both roots and shoots (6.9- and 101-fold, respectively). Collectively, these expression patterns resemble those in rice (Kobayashi et al. 2014), suggesting that sorghum mainly relies on strategy II for Fe uptake using Fe deficiency-inducible genes, similarly to rice.

Sequence and enzymatic activity of *SbNAS1*

NAS is a key enzyme in strategy II plants because of increased demand for NA and MAs in roots under Fe-deficient conditions. Plant NAS proteins are phylogenetically divided into two classes, Class I and Class II (Mizuno et al. 2003; Zhou et al. 2013; Fig. 2a). Among the three NAS proteins in sorghum, *SbNAS1* belongs to Class I, while *SbNAS3a*, and *SbNAS3b* belong to Class II. *SbNAS1* was deduced to be the main NAS protein in sorghum, contributing to the synthesis of MAs for Fe uptake from the rhizosphere because this function is generally attributed to Class I NAS (Inoue et al. 2003; Mizuno et al. 2003), and this gene was the only sorghum NAS gene strongly upregulated in Fe-deficient roots (Fig. 1a).

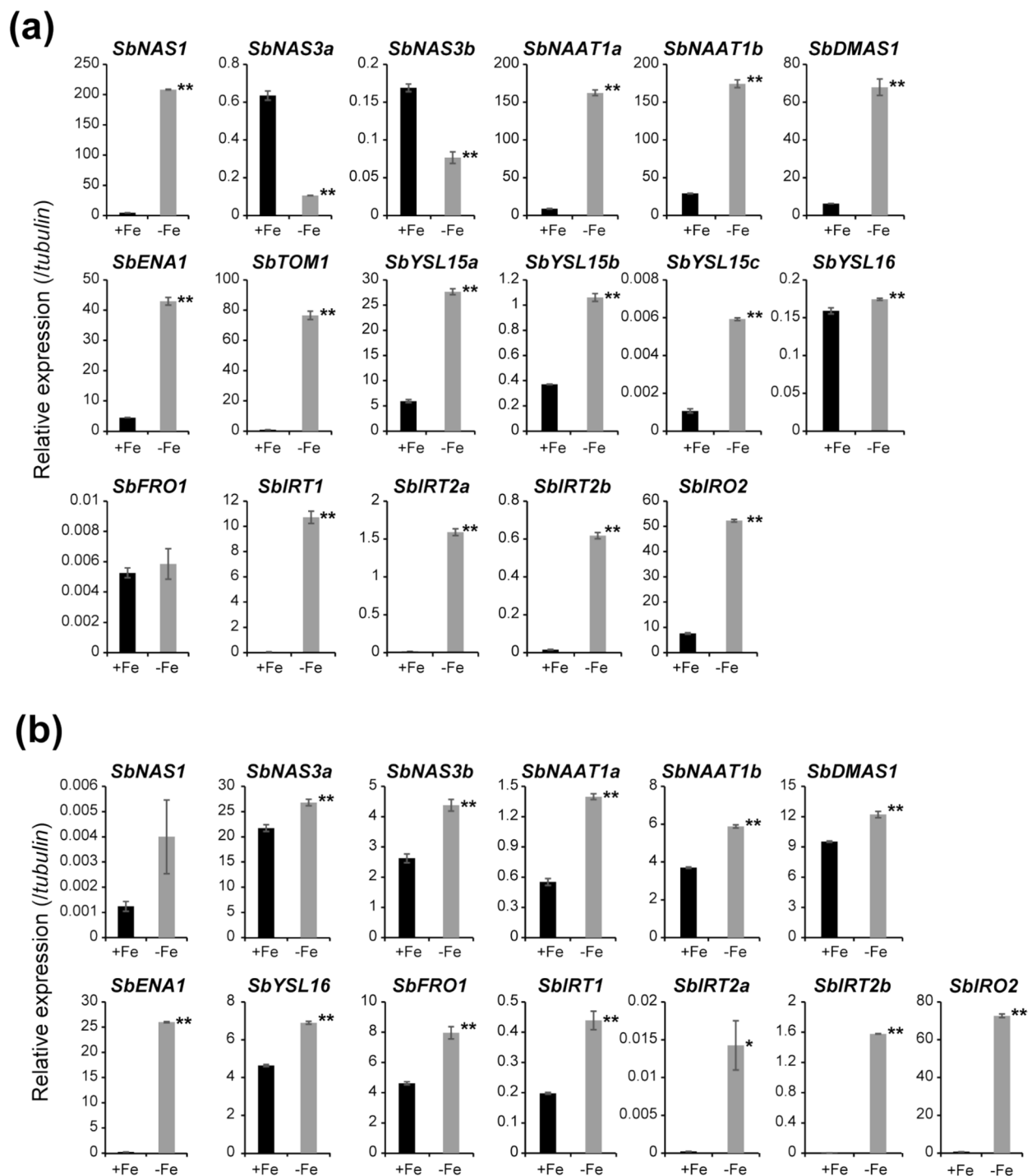


Fig. 1 Expression profiles of Fe uptake-related genes in roots **(a)** and shoots **(b)** of sorghum. Fifteen-day-old sorghum seedlings grown hydroponically were cultured under Fe sufficiency (+Fe) or deficiency (–Fe) for 3 days. Relative transcript level of each gene listed in Table S2 was determined by RT-qPCR and shown as the normalized expression level relative to a tubulin gene (*Sb09g004420*) for both (a)

and (b). Results are presented as the mean $\pm$ SD of three replicates and the significance of differences was analyzed by Student's *t* test (* $P < 0.05$; ** $P < 0.01$). The expression of *SbTOM1*, *SbYSL15a*, *SbYSL15b*, and *SbYSL15c* in shoots was under the detection limit regardless of the Fe status

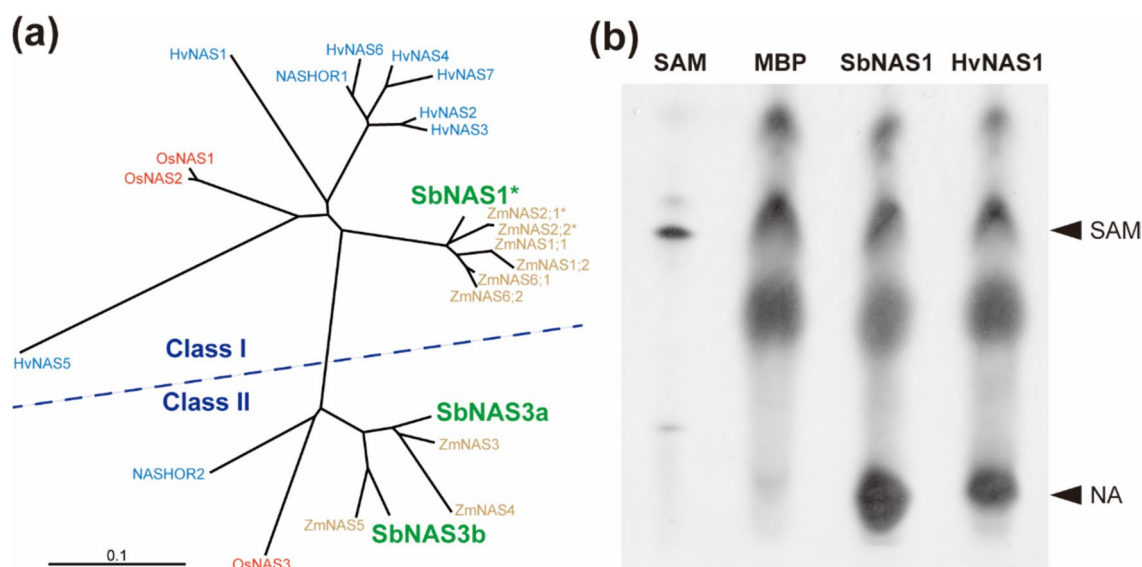


Fig. 2 Analysis of NAS homologs in sorghum. **a** Unrooted phylogenetic tree of NAS in sorghum, rice, maize, and barley. The proteins and their accession numbers used for alignment and phylogenetic tree construction are as follows: ZmNAS1;1 [GRMZM2G385200], ZmNAS1;2 [GRMZM2G312481], ZmNAS2;1 [GRMZM2G030036], ZmNAS2;2 [GRMZM2G124785], ZmNAS3 [GRMZM2G478568], ZmNAS4 [GRMZM2G439195], ZmNAS5 [GRMZM2G050108], ZmNAS6;1 [GRMZM2G704488], and ZmNAS6;2 [AC233955.1_FGT003] from maize (*Zea mays*); NASHOR1 [AF136941], NASHOR2 [AF136942], HvNAS1 [AB010086], HvNAS2 [AB011265], HvNAS3 [AB011264], HvNAS4 [AB011266],

HvNAS5 [AB011268], HvNAS6 [AB011269], and HvNAS7 [AB019525] from barley (*Hordeum vulgare*); and OsNAS1 [AK112069], OsNAS2 [AK112011], and OsNAS3 [AK070656] from rice (*Oryza sativa*). Asterisks indicate NAS proteins with a tandem repeat structure like SbNAS1. **(b)** Confirmation of NAS activity of SbNAS1. Purified MBP, MBP-HvNAS1, and MBP-SbNAS1 were incubated with 20 μ M [14 C]S-adenosyl-L-methionine (SAM) at 25 $^{\circ}$ C for 20 min, after which the products were analyzed by thin-layer chromatography. Arrowheads indicate SAM and NA

SbNAS1 had a rare tandem repeat structure (Fig. S2a). To clarify the enzymatic activity of SbNAS1, recombinant SbNAS1 fused to MBP at the N-terminus was expressed in *E. coli* and purified. MBP-HvNAS1 was used as a positive control (Higuchi et al. 1999; Mizuno et al. 2003). Similar to the findings in the control, MBP-SbNAS1 clearly displayed NAS activity (Fig. 2b). These results suggest that SbNAS1 is the principal NAS in sorghum contributing to the biosynthesis of MAs in roots for strategy II-based Fe uptake.

Secretion of MAs and ferric-chelate reductase activity in sorghum roots

To further characterize possible Fe uptake mechanisms in sorghum roots, secretion of MAs from the sorghum roots was analyzed after Fe-sufficient and -deficient hydroponic culture (Fig. 3a, b). In the HPLC chromatogram from sorghum root washings, we detected three peaks attributable to MAs (Fig. 3a, peaks 1–3), which was further verified by their disappearance when hypochlorite was omitted in the detection (Fig. 3a; Mori et al. 1987; Mori and Nishizawa 1987; Nozoye et al. 2017). Among these, peak 3 was deduced as DMA by comparison of the retention time with that of DMA standard and co-chromatograph of the root-washing

with DMA standard (Fig. 3a). Some difference in retention time may be due to a matrix effect. Peaks 1 and 2 were deduced to be epiHDMA and 3-hydroxy-2'-deoxymugineic acid (HDMA), respectively, by co-chromatography with root washings from *Brachypodium distachyon* (Fig. S3; Yordem et al. 2011), which also confirmed that peak 3 corresponds to DMA. The secreted amount of DMA was increased to 13 times under Fe deficiency, but was lower than that secreted from similarly cultured Fe-deficient rice (Fig. 3b). Similar induction of secretion of other MAs was also observed under Fe deficiency (Fig. S3).

We also measured ferric-chelate reductase activity on the sorghum root surface. The reductase activity was lower than that of typical Strategy I plants (Holden et al. 1991; Marschner 1995), and was significantly decreased to 34% by Fe deficiency treatment (Fig. 3c). This trend is similar to that in rice (Ishimaru et al. 2006).

Intercropping with rice or barley plants with high MA secretion improves Fe deficiency symptoms of sorghum

To assess whether the susceptibility of sorghum plants to Fe deficiency is attributable to its low capacity for

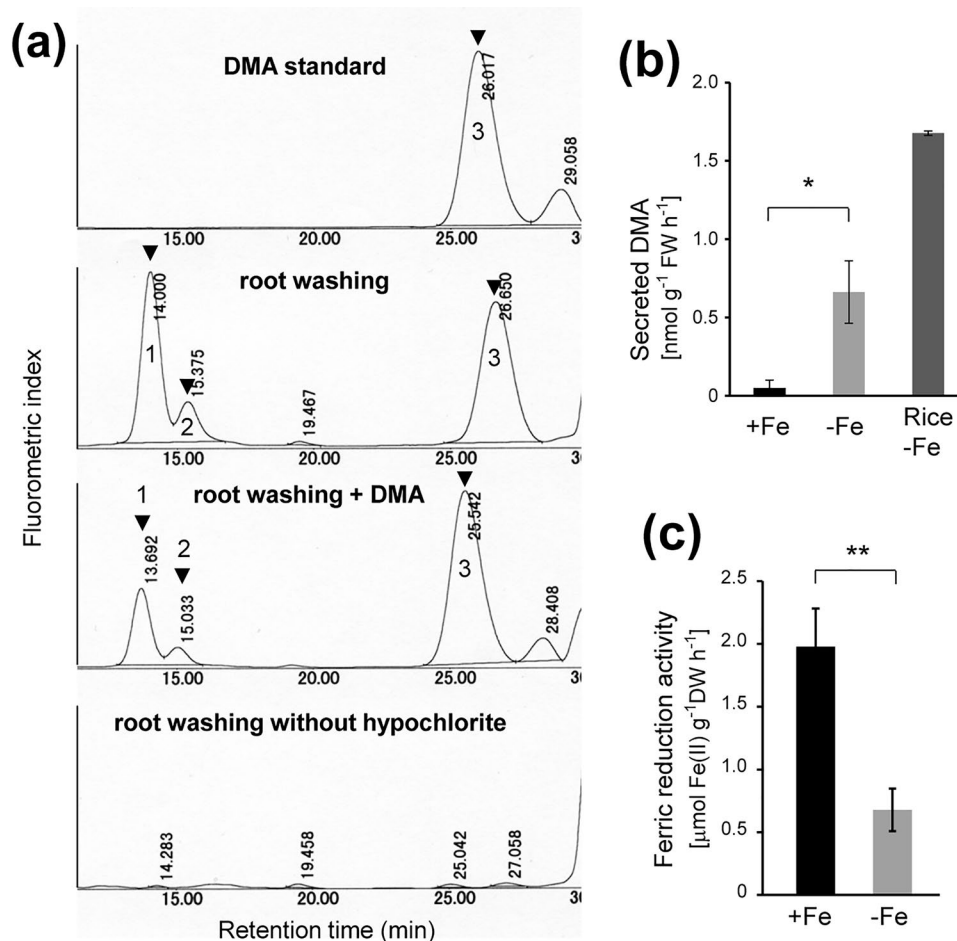


Fig. 3 Secretion of mugineic acid family phytosiderophores (MAs) and ferric-chelate reductase activity in sorghum roots. **a** Chromatograms of MAs by high-performance liquid chromatography. “**DMA standard**”: 2'-deoxymugineic acid (DMA) standard solution. “**root washing**”: Root washing of sorghum seedling cultured under Fe deficiency for 7 days. “**root washing + DMA**”: Co-chromatography of the above two samples. “**root washing without hypochlorite**”: The above root washing detected without hypochlorite which is required for detection of secondary and tertiary amines including MAs (Mori et al. 1987). Peaks 1 and 2 were deduced to be 3-epihydroxy-2'-deoxymugineic acid (epiHDMA) and 3-hydroxy-2'-deoxymugineic acid (HDMA), respectively (Fig. S3). Peak 3 was deduced to be DMA. **b** Secreted DMA amounts. Root washings were

collected from sorghum seedlings cultivated hydroponically under Fe sufficiency (+Fe) or deficiency (–Fe) for 7 days, as well as rice seedlings cultivated hydroponically under Fe deficiency for 7 days (“Rice–Fe”). Results are presented as the mean $\pm$ SD of four replicates of three plants each. FW: fresh weight. Significant difference compared to +Fe was analyzed by Mann–Whitney U-test (* $P < 0.05$). **c** Ferric-chelate reductase activity on the root surface. Sorghum seedlings were treated hydroponically with (+Fe) or without Fe (–Fe) for 4 days. Results are presented as the mean $\pm$ SD of four independent experiments and the significance of differences was analyzed by Student's t test (** $P < 0.01$). DW: dry weight

MA secretion, we conducted intercropping analysis of sorghum with rice and barley (Fig. 4; Fig. S4). Sorghum plants were hydroponically cultured under Fe deficiency without other plants (Mono), or with rice or barley plants (Inter) in the same hydroponic box (Fig. S4). We used two rice lines: non-transformant (NT) and a transgenic line carrying barley genomic fragment of *IDS3*, which enables the enhanced production of MA and DMA (Kobayashi et al. 2001). Intercropping of sorghum with these rice lines slightly mitigated the appearance of chlorosis in the new

leaves, quantified as SPAD value (relative chlorophyll concentration; Fig. 4b). Also, root growth tended to be better with the *IDS3* line (Fig. 4a). Moreover, intercropping of sorghum with barley strongly rescued the chlorosis in the sorghum leaves (Fig. 4a, b). This effect was still evident at 28 days after treatment onset (Fig. S4). These results indicate that the supply of large amounts or more types of MAs is effective in mitigating sorghum Fe deficiency.

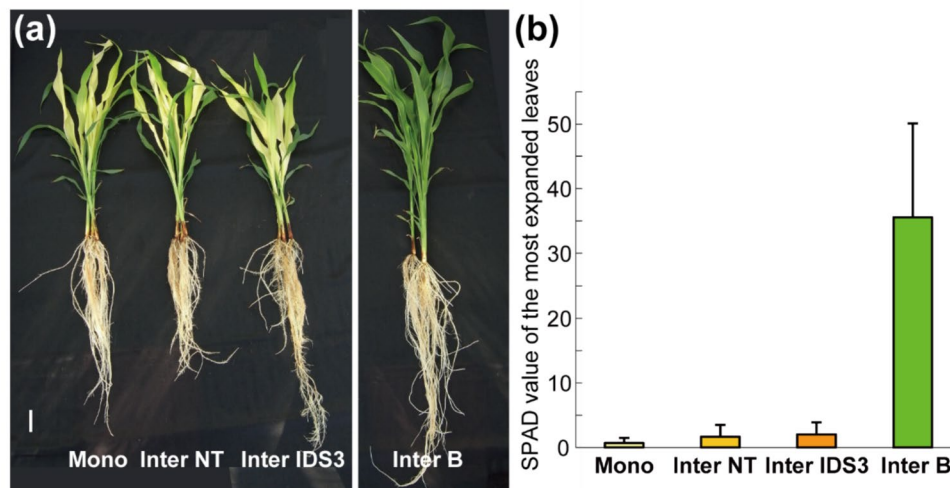


Fig. 4 Intercropping of sorghum with rice or barley under Fe-deficient hydroponic culture. **a** Appearance of sorghum after intercropping with rice for 12 days under Fe deficiency. Mono: monocropping. Inter NT: intercropping of 3 sorghum plants with 30 non-transformant rice plants. Inter IDS3: intercropping of 3 sorghum plants with

30 rice plants carrying barley *IDS3* genomic fragment. Inter B: intercropping of 3 sorghum plants with 30 barley plants. Scale bar represents 5 cm. **b** The Soil Plant Analysis Development (SPAD) value of the most expanded leaves after intercropping for 12 days. The mean ± SD is shown (n=3)

Transgenic sorghum plants carrying quadruple gene cassettes show improved MA secretion, Fe uptake and tolerance to low Fe availability

We then designed and constructed a quadruple expression vector for the transformation of sorghum to confer tolerance of Fe deficiency (Fig. 5a; Fig. S1). Based on the results presented above, we aimed to improve the expression of genes for strategy II-based Fe acquisition by overexpressing rice transcription factor *OsIRO2* (35S promoter-*OsIRO2* cassette; Ogo et al. 2007, 2011), enhancing NA and MA production via a barley *HvNAS1* genomic fragment (Higuchi et al. 2001a, b), boosting MA production via a barley *IDS3* genomic fragment (Kobayashi et al. 2001), and introducing exogenous ferric-chelate reductase activity in Fe-deficient roots by *Refre1/372* (Oki et al. 2004; Ishimaru et al. 2007) driven by the newly subcloned sorghum *SbIRT1* promoter. Although an efficient transformation system of sorghum has not yet been developed, we successfully obtained one line transformed with this quadruple vector, designated as “line N22,” along with one line for vector control, designated as “VC.” Southern hybridization analysis using the hygromycin resistance gene (*HPT*) in the transgene as a probe detected a single band in the genome of the line N22 (Fig. S5), suggesting the presence of the transgene in a single copy. In addition, genomic PCR analysis successfully amplified the concatenated sequence of *OsIRO2* and *Refre1/372* expression cassettes in four T_2 progenies of N22, but not in VC progenies (Fig. S5).

We hydroponically cultured T_1 plants of N22 and VC without Fe supplementation for 18 days. The VC showed severe chlorosis in new leaves, but N22 showed strong Fe deficiency tolerance, with most leaves retaining a deep green color (Fig. 5b). Moreover, the line N22 secreted markedly enhanced amounts of MAs, with an increased ratio of 18 times in total (Fig. 5c). Furthermore, N22 roots secreted MA, which was absent in VC root washings (Fig. 5c; Fig. S6), and the amount of secretion of unidentified MAs was much higher than that of DMA in the line N22, in contrast to the scarce secretion of these MAs in the VC (Fig. 5c; Fig. S6). These effects are deduced to be facilitated by 2'-hydroxylation of MAs catalyzed by the expression of barley *IDS3*. In addition, the ferric-chelate reductase activity of the N22 roots was moderately increased compared with that of VC (Fig. 5d), suggesting the expression and function of introduced *Refre1/372*.

We also measured the metal concentrations in the roots and leaves of these T_1 plants after Fe-sufficient or -deficient hydroponic culture (Fig. 6). The line N22 showed higher Fe concentrations than VC in both young leaves and roots, with significant differences under Fe sufficiency (Fig. 6a, b). The line N22 also showed lower Zn, Mn, and Cu concentrations in Fe-sufficient roots, and higher Mn concentration in Fe-deficient roots (Fig. 6b).

We further examined the tolerance of the transformants to low Fe availability in calcareous soil pots. Similar to the hydroponics, the line N22 retained greener leaves, while VC suffered from interveinal chlorosis in calcareous soil pots

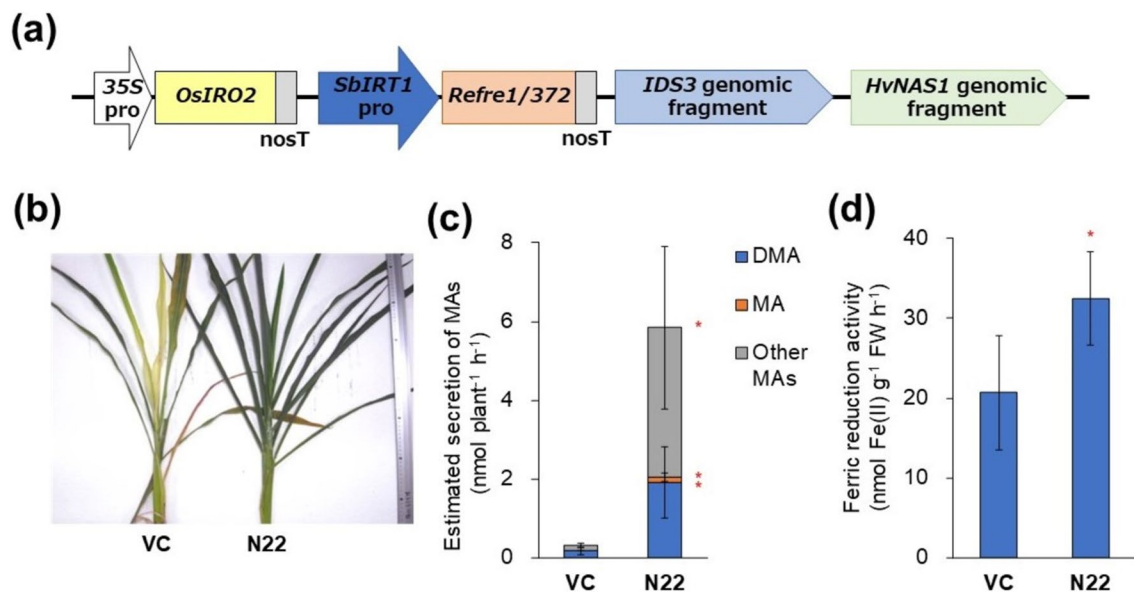


Fig. 5 The vector structure, Fe deficiency tolerance, secretion of mugenic acid family phytosiderophores (MAs), and ferric-chelate reductase activity in transgenic sorghum. **a** The vector structure of the “quadruple” sorghum and rice lines. The binary vector contained the depicted four cassettes, along with hygromycin and kanamycin resistance genes and 35S promoter- β -glucuronidase (*GUS*) gene with an intron, inside the T-DNA region of the pBIGRZ1 vector backbone (Fig. S1; Akiyama et al. 1997; Kobayashi et al. 2001). 35S pro, cauliflower mosaic virus 35S promoter; *OsIRO2*, rice iron-related transcription factor 2; nosT, *Agrobacterium tumefaciens* nopaline synthase terminator; *SbIRT1* promoter, sorghum iron-regulated transporter 1 promoter; *Refre1/372*, reconstructed yeast ferric-chelate reductase 1 variant 372; *IDS3*, barley iron deficiency-specific clone 3; *HvNAS1*, barley nicotianamine synthase 1. The vector is not drawn to scale. **b** Plant appearance of T₁ plants at day 18 of Fe-deficient hydro-

ponic culture. VC, vector control; N22, quadruple line. **c** Estimated amounts of secretion of MAs from T₁ plants at day 18 of Fe-deficient hydroponic culture, based on the chromatogram shown in Fig. S6. Amounts were calculated by the comparison of peak areas with DMA, MA, and epiHMA standards. The amounts of “other MAs” were estimated using the epiHMA standard as a proxy for the peaks with similar retention times, such as epiHDMA, and HDMA. Results are presented as the mean $\pm$ standard error (SE) (n=4 for VC; n=3 for N22). The significance of differences compared to VC was analyzed by Mann–Whitney U-test (**P* < 0.05). **d** Ferric-chelate reductase activity on the root surface of T₁ plants at day 14 of Fe-deficient hydroponic culture. Results are presented as the mean $\pm$ SD (n=5 for VC; n=7 for N22). DW: dry weight. The significance of differences was analyzed by Student’s *t* test (**P* < 0.05)

(Fig. 7a, b). The line N22 also showed higher SPAD values in the newest leaves than line VC at many time points during 17–47 days after transplanting in T₁ generation (Fig. 7c) and during 35–50 days after transplanting in T₂ generation (Fig. 7d). These two lines showed similar shoot heights during the treatment, with slightly greater height in N22 at days 1 and 3 but not thereafter in the T₁ generation (Fig. S7). These results suggest that the introduction of the quadruple gene cassettes is effective for improving MA secretion, Fe uptake and preserving leaf chlorophyll during low Fe availability.

Transgenic rice plants carrying the quadruple cassettes show improved tolerance to low Fe availability and seed Fe accumulation

To further confirm the efficacy of the quadruple cassettes for conferring tolerance of Fe deficiency, rice was transformed with the same vector (Fig. 5a). Twelve transformants (designated as Q lines for quadruple) were obtained, in

which the presence of the four transgenes was confirmed (Fig. S8a), among which nine lines were screened for tolerance to low Fe availability in calcareous soil in the T₁ generation (Fig. S8b, c). Four tolerant lines, Q1, Q7, Q8, and Q9, were selected for further analysis. All of these lines showed strong tolerance in calcareous soil pots again in the T₂ generation, with higher SPAD values of the newest leaves and higher shoots than NT (Fig. S9). The Q1, Q7, and Q8 lines also produced more tillers and higher panicle and straw biomass than NT (Fig. S9). In addition, the Q9 line exhibited the ability to secrete MA, and a higher amount of DMA than NT (Fig. S10a), similar to rice lines with the introduction of *IDS3* (Kobayashi et al. 2001). This line also showed increased ferric-chelate reductase activity at the root surface than NT (Fig. S10b). These results suggest the successful expression and function of introduced gene cassettes, including *SbIRT1* promoter, leading to strong tolerance in calcareous soil.

We further compared the traits of the quadruple rice lines Q1, Q7, and Q8 at the T₃ generation with previously

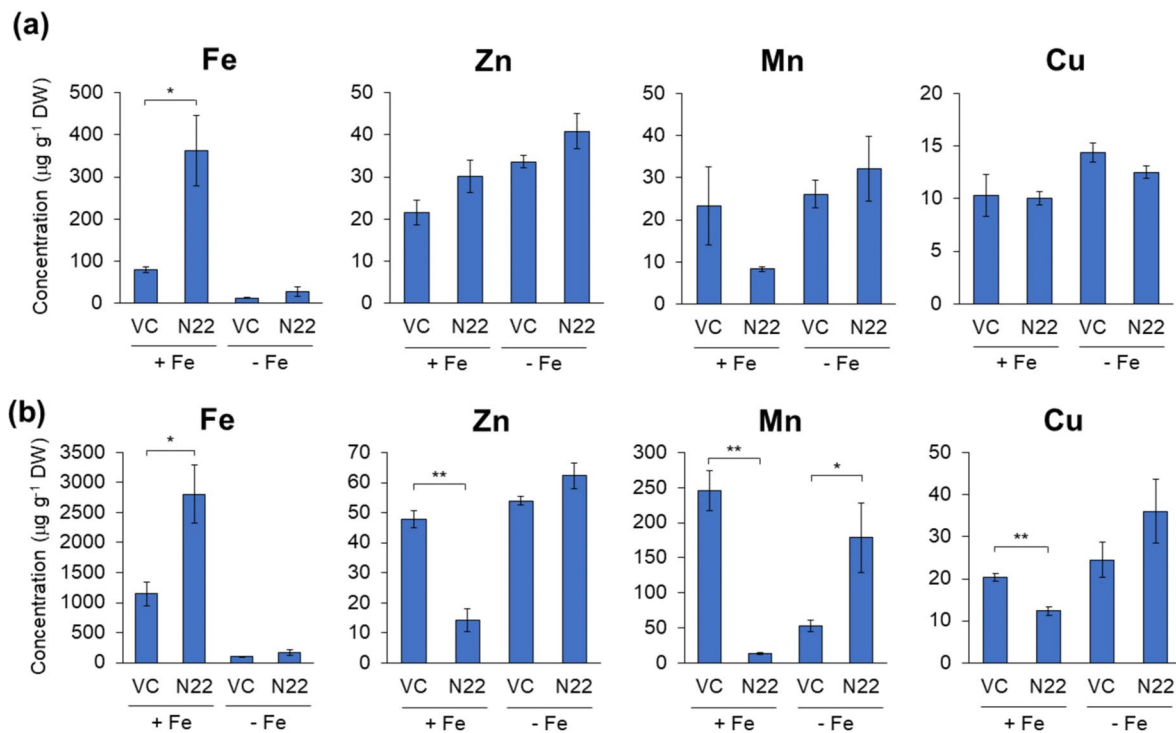


Fig. 6 Metal concentrations in T₁ plants of transgenic sorghum. **(a)** Youngest leaves. **(b)** Roots. VC, vector control; N22, quadruple line. Transgenic sorghum plants were hydroponically grown under Fe-sufficient (+Fe) or -deficient (-Fe) condition for 10 days. Results are

presented as the mean $\pm$ standard error (SE) (n=4 for +Fe VC shoots and +Fe N22 roots; n=7 for -Fe N22 shoots and roots; n=5 for others). DW: dry weight. The significance of differences was analyzed by two-sample Student's t-test (* P <0.05; ** P <0.01)

characterized lines of double cassettes (designated as line D; 35S promoter-*OsiRO2* + *OsiRT1* promoter-*Refre1/372*; Masuda et al. 2017), and triple cassettes (designated as line T; 35S promoter-*OsiRO2* + *OsiRT1* promoter-*Refre1/372* + *IDS3* genomic fragment; Masuda et al. 2019). During cultivation in calcareous soil pots, all of these transgenic lines showed superior SPAD values and shoot heights than NT, to different degrees (Fig. 8a–c); line Q7 tended to show the highest SPAD value, while line Q8 tended to show the lowest SPAD values and shoot height, among the transgenic lines. Similar trends were also observed for panicle and straw biomass at seed maturation, with significantly higher yields than in NT, with the exception of panicle yield of line Q8 (Fig. 8d). In addition, lines Q1, and Q7 showed significant increases in Fe and Zn concentrations in brown seeds compared with NT, while the other lines showed similar but weaker tendencies (Fig. 8e, f). These results suggest that the introduction of the quadruple gene cassettes is effective for conferring tolerance of Fe deficiency and Fe accumulation in rice.

Discussion

In the present study, we aimed to improve sorghum traits under low Fe availability. We first identified putative genes for sorghum Fe uptake and translocation and examined their expression. Sorghum possessed almost all gene counterparts, which play important roles in rice for Fe acquisition (Table S2; Fig. S2). In addition, their transcriptional responses to Fe status (Fig. 1) were largely similar to those in rice, barley, and maize (Suzuki et al. 2006; Kobayashi et al. 2014; Wairich et al. 2019; Zhang et al. 2022). Previously, Wairich et al. (2019) reported similar induction of *SbIRO2* and *SbYS1* (which could correspond to *SbYSL15a*, *SbYSL15b*, or *SbYSL15c* in our analysis) in Fe-deficient roots, while they reported no induction of *SbIRT1*, possibly because of a shorter duration of treatment with Fe deficiency. These results suggest that molecular mechanisms for Fe homeostasis

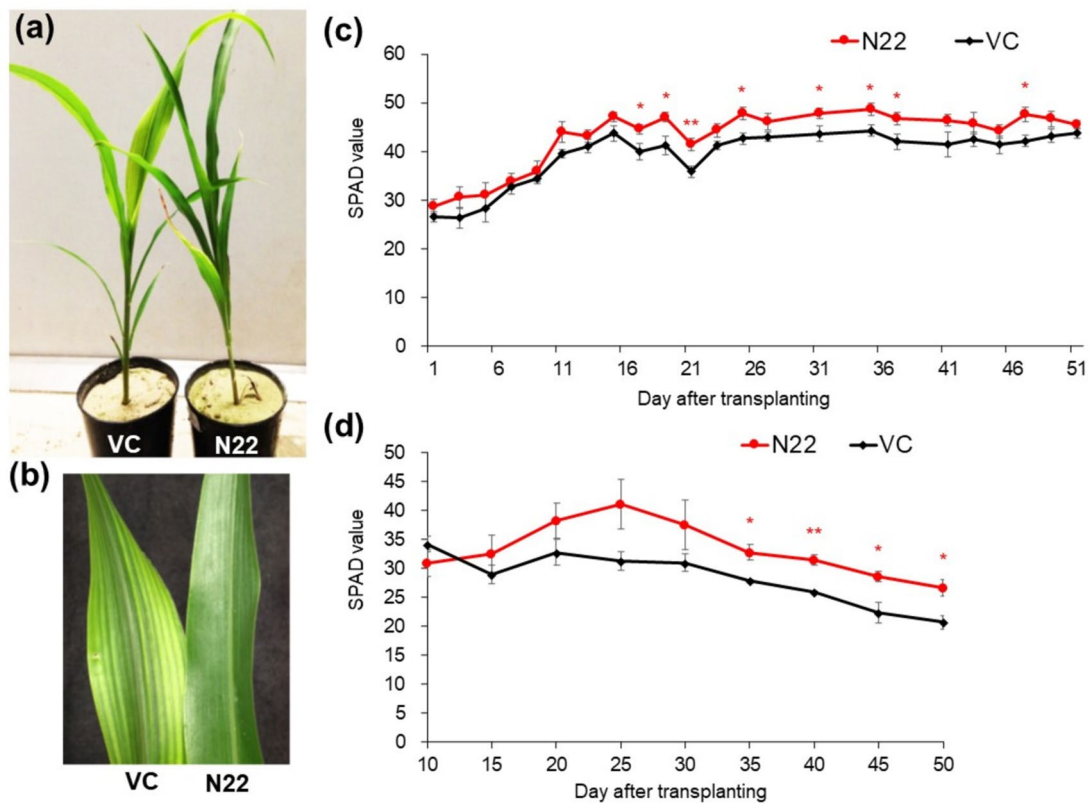


Fig. 7 Features of the transgenic sorghum lines under pot cultivation in calcareous soil. **(a)** Representative T_1 plants on day 17 after transplanting. **(b)** Enlarged image of leaves in **(a)**. **(c)**, **(d)** The Soil Plant Analysis Development (SPAD) value of the newest leaves at T_1 **(c)** and T_2 **(d)** generations. VC, vector control; N22, quadruple

line. **(c)** and **(d)** show the mean $\pm$ standard error (SE) [$n=9$ each for **(c)** and $n=3$ each for **(d)**]. Asterisks indicate significant differences compared to the NT at each time point (two-sample Student's t -test; * $P < 0.05$; ** $P < 0.01$)

are largely conserved between rice and sorghum, and our knowledge on the enhancement of rice tolerance of Fe deficiency could be also applicable to sorghum.

Sorghum secreted three kinds of MAs, among which the major ones were deduced to be DMA and epiHDMA (Fig. 3, Fig. S3). DMA is biosynthesized in all of the graminaceous plants tested, while epiHDMA is deduced to be biosynthesized from DMA in limited species by hydroxylation at the 3-position by IDS2 dioxygenase (Nakanishi et al. 2000; Ueno et al. 2007). Although a gene similar to barley *IDS2* was found in sorghum (Sb08g016400; Fig. S2j), this gene was not induced by Fe deficiency in roots (data not shown), irrespective of the induced secretion of possible epiHDMA and HDMA under Fe deficiency (Fig. S3). There was no homolog closer to *IDS3* than Sb08g016400 in sorghum (Fig. S2j), consistent with the absence of MA secretion in sorghum (Fig. 3; Fig. S3; Fig. S6). EpiHDMA could also correspond to “unknown phytosiderophore” secreted from sorghum roots (Römhelt and Marschner 1990), although its retention time appeared to be longer than that of epiHDMA in recent analyses including the present results (Fig. S3, Ueno et al. 2007; Yordem

et al. 2011). The amount of secreted DMA was increased by Fe deficiency (Fig. 3b), consistent with the substantial upregulation of *SbNAS1*, *SbNAAT1a*, *SbNAAT1b*, *SbDMAS1*, and *SbTOM1* in roots by Fe deficiency (Fig. 1a). However, the amount of DMA (Fig. 3b) was very low, consistent with previous reports showing much lower secretion of MAs from sorghum compared with that in other graminaceous plants such as barley, rye, wheat, and maize (Kawai et al. 1988; Römhelt and Marschner 1990; Higuchi et al. 1996). Higuchi et al. (1996) also reported still lower secretion of MAs per plant in rice than in sorghum, while our analysis showed higher MA secretion per fresh weight in rice than in sorghum (Fig. 3b), presumably due to differences in cultivars, plant sizes and growth conditions. Rice is well known to be highly susceptible to Fe deficiency, mainly because of the low capacity for MA secretion (Mori et al. 1991). Indeed, tolerance of Fe deficiency among graminaceous plants is thought to be correlated with the amount of MAs secreted. In general, the highest MA secretion is observed in barley, followed by wheat, rye, maize, and then sorghum (Römhelt and Marschner 1990). Meanwhile, sorghum has sufficient capacity to take up supplemented Fe(III)-MAs, similar to

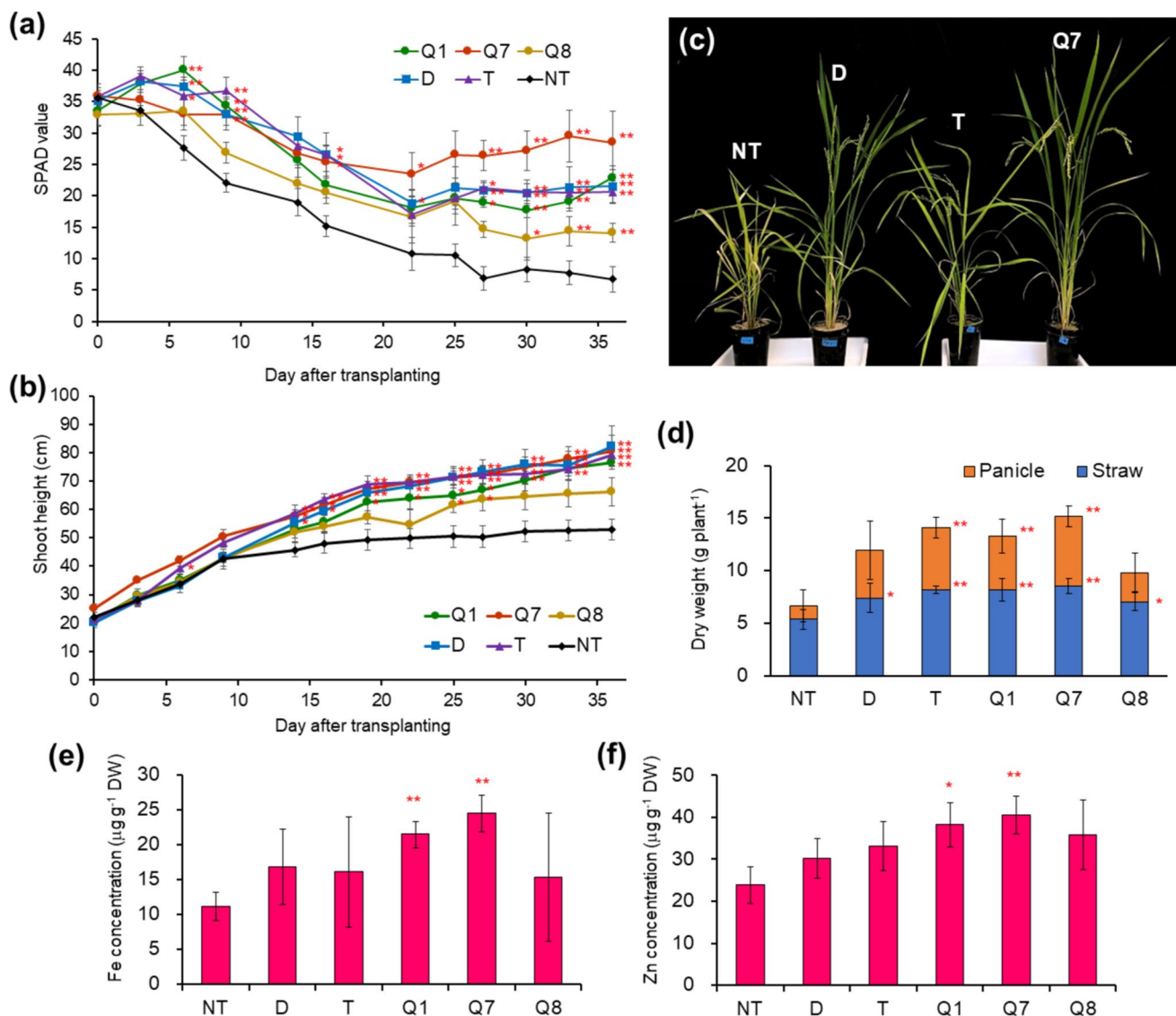


Fig. 8 Comparison of the quadruple T₃ transgenic rice lines with double and triple transgenic rice lines and non-transformants under pot cultivation in calcareous soil. **a** The Soil Plant Analysis Development (SPAD) value of the newest leaves. **b** Shoot height. **c** Representative plants on day 59 after transplanting. **d** Panicle and straw dry weights after seed maturation. **e**, **f** Fe (**e**) and Zn (**f**) concentrations in brown seeds. Q1, Q7, Q8: quadruple lines introduced with pBI-GRZ-*OsIRO2-Refre1-IDS3-HvNAS1* (Fig. 5a). D: double line introduced with 35S promoter-*OsIRO2* + *OsIRT1* promoter-*Refre1/372*

(Masuda et al. 2017). T: triple line introduced with 35S promoter-*OsIRO2* + *OsIRT1* promoter-*Refre1/372* + *IDS3* genomic fragment (Masuda et al. 2019). NT: non-transformants. **(a)** and **(b)** show mean $\pm$ SE ($n=5$). **d–f** show mean $\pm$ SD [$n=3$ for NT in **(e)** and **(f)**; $n=4$ for Q1 and D lines in **(e)** and **(f)**; $n=5$ for the others]. Asterisks indicate significant differences compared to the NT at each time point or organ (two-sample Student's *t*-test; * $P<0.05$; ** $P<0.01$)

other graminaceous plants (Römhelt and Marschner 1990). Moreover, all of the analyzed genes for strategy II showed responsiveness to Fe deficiency in roots, shoots, or both (Fig. 1), suggesting that the response to Fe deficiency in sorghum is functional. Furthermore, the symptoms of Fe deficiency in sorghum were rescued by cocultivation with barley or rice with a high capacity to secrete MAs (Fig. 4). These facts suggested that sorghum has a bottleneck in the step of biosynthesis and/or secretion of MAs.

To facilitate the secretion of MAs in sorghum, the potency of NA synthesis in roots is an important factor because NA is the exclusive precursor of MAs. The amount of MA secretion is correlated to NAS activity in the roots of graminaceous plants (Higuchi et al. 1996). The introduction of *HvNAS1* into rice enhances the NAS activity in roots and DMA secretion (Higuchi et al. 2001a). Generally, NAS proteins belonging to Class I in graminaceous plants are strongly induced by Fe deficiency in roots, providing NA

for the synthesis of MAs. In sorghum, SbNAS1 was the only NAS belonging to Class I (Fig. 2a). Consistent with this, *SbNAS1* was strongly induced by Fe deficiency in the roots (Fig. 1a). As in sorghum, rice has only three NAS proteins, among which OsNAS1, and OsNAS2 belong to Class I, both of which are strongly induced by Fe deficiency (Higuchi et al. 2001b; Inoue et al. 2003). In maize and barley, there are six, and eight NAS proteins in Class I, respectively (Fig. 2a). These findings indicate that sorghum has fewer Class I NAS genes than other major cereal crops. Moreover, SbNAS1 has a rare tandem repeat structure (Fig. S2a). NAS with such a structure was identified only in sorghum and maize (Mizuno et al. 2003; Zhou et al. 2013). Although SbNAS1 showed NAS activity (Fig. 2b), it is not known whether such tandem repeat structure affects NA synthesis *in planta*. NAS proteins belonging to Class II are induced by Fe excess in shoots, and are thought to be involved in internal metal translocation and detoxification of excess Fe (Inoue et al. 2003; Zhou et al. 2013; Aung et al. 2019). Consistent with this, the expression of *SbNAS3a* and *SbNAS3b* belonging to Class II was lower in roots than in shoots and downregulated by Fe deficiency in roots (Fig. 1a, b), suggesting that the contribution of these genes to strategy II Fe uptake is limited. Thus, we hypothesized that the introduction of an additional NAS gene into sorghum would increase the tolerance of Fe deficiency mediated by the increased secretion of MAs. Indeed, the introduction of a barley *HvNAS1* genome fragment including the promoter region into rice improved its growth in calcareous soil under field conditions (Suzuki et al. 2008).

Rice does not synthesize MA because it lacks the MA synthase gene *IDS3*, which converts DMA into MA (Kobayashi et al. 2001). Fe(III)-MA is more stable than Fe(III)-DMA under slightly acidic conditions (von Wirén et al. 2000). In barley, *IDS3* expression is strongly induced by Fe deficiency (Nakanishi et al. 1993). Transgenic rice lines carrying a barley *IDS3* genome fragment exhibit the ability to secrete MA from the roots, in addition to enhanced DMA secretion and improved tolerance in calcareous soil fields (Kobayashi et al. 2001; Suzuki et al. 2008). Similar to rice, sorghum plants secreted DMA, but not MA (Fig. 3a, b; Fig. S6). Thus, we deduced that the introduction of *IDS3* into sorghum would facilitate MA production and improve the growth in calcareous soil.

In nongraminaceous plants, expression of the Fe^{2+} transporter IRT is induced by Fe deficiency coincidentally with ferric-chelate reductase activity in roots, which enables the efficient uptake of Fe from the soil (Connolly et al. 2003). Because Fe^{2+} is easily oxidized to Fe^{3+} and precipitated, IRT cannot effectively absorb Fe^{2+} from the soil without ferric-chelate reductase activity on the root surface in aerobic conditions. The expression of *SbIRTs* was strongly induced by Fe deficiency in roots (Fig. 1a). However, ferric-chelate reductase activity in sorghum roots

was decreased by Fe deficiency (Fig. 3c). Consistent with this, sorghum did not have *FRO* gene expression induced in roots by Fe deficiency (Fig. 1a), similar to the findings in rice (Ishimaru et al. 2006). Previously, we reported that the introduction of an artificial ferric-chelate reductase gene, *Refre1/372*, under the control of the *OsIRT1* promoter in rice strongly improved the growth in calcareous soil (Ishimaru et al. 2007). Thus, we designed and constructed a new cassette carrying *SbIRT1* promoter-*Refre1/372* for introduction into sorghum. In addition, we combined another cassette for overexpression of the transcription factor *OsIRO2*, which conferred tolerance in calcareous soil in rice (Ogo et al. 2011), which is more effective when combined with the *OsIRT1* promoter *Refre1/372* and barley *IDS3* genomic fragment (corresponding to the D and T lines in Fig. 8; Masuda et al. 2017, 2019).

Based on these considerations, we constructed four expression cassettes: *35S* promoter-*OsIRO2*, *SbIRT1* promoter-*Refre1/372*, and barley *IDS3* and *HvNAS1* genomic fragments (Fig. 5a), in a single transformation vector. This construct was introduced into sorghum, as well as rice for proof-of-concept in a well-analyzed biological system. We used the rice *OsIRO2* gene and the sorghum *SbIRT1* promoter for both sorghum and rice transformation because of the high conservation of transcription factors and related *cis*-acting elements for Fe deficiency responses among plant species, and their successful function in many heterogeneous systems (Higuchi et al. 2001b; Kobayashi et al. 2001, 2003; Ogo et al. 2006; Ito et al. 2009). We obtained one sorghum transformation line, N22, which showed improved chlorophyll concentrations in new leaves under Fe deficiency in both hydroponics (Fig. 5b) and calcareous soils (Fig. 7). This line also showed a marked increase in secretion of MAs (Fig. 5c), higher ferric-chelate reductase activity (Fig. 5d), and increased Fe accumulation in roots and leaves (Fig. 6). Insertion of *OsIRO2* and *Refre1/372* cassettes, in addition to hygromycin resistance gene, was confirmed in this line (Fig. S5). Notably, line N22 secreted MA, which was absent from VC (Fig. 5c; Fig. S6) and wild-type sorghum (Fig. 3b). This result strongly suggested that the phenotype of line N22 is conferred by the introduced barley *IDS3* dioxygenase, rather than disturbance of an endogenous system by a possible unexpected gene mutation. Line N22 also secreted a higher proportion of unidentified MAs compared with DMA, unlike VC (Fig. 5c; Fig. S6) or wild-type sorghum (Fig. 3b; Fig. S3). This result suggests further hydroxylation of MA produced by exogenous *IDS3* in line N22. Indeed, 3-hydroxylation of MA to produce epiHMA or HMA has been deduced to be catalyzed by *IDS2* dioxygenase (Nakanishi et al. 2000), similar to 3-hydroxylation from DMA to epiHDMA or HDMA, which is suggested to be present in wild-type sorghum (Fig. 3b; Fig. S3). Thus, the introduction of barley *IDS3* is thought to facilitate further hydroxylation of MAs,

improving both the amounts and the kinds of MAs produced, presumably further aided by enhancement of the pathway of DMA production by *OsIRO2* and *HvNAS1* transgenes. Similarly, *IDS3*-introduced rice lines not only secrete MA, but also increase total production of MAs (Kobayashi et al. 2001), as also observed in our Q9 rice line (Fig. S10a). Such reinforcement of the strategy II system, together with the introduction of Fe deficiency-inducible ferric-chelate reductase (Fig. 5d), could have contributed to the enhanced tolerance of Fe deficiency shown by line N22. In summary, enhancement of Fe uptake and improved leaf color in calcareous soil in the line N22 is considered to be caused by combined effects of *OsIRO2*, *Refre1/372*, *IDS3* and presumably *HvNAS1* transgenes. Because of technical limitations, we were unable to obtain more than one line of the sorghum transformants. Also, the line N22 did not show improved shoot height during the cultivation in calcareous soils (Fig. S7). Precise demonstration of the transgene effects on improvement in sorghum biomass production is an important issue for the future.

As an alternative method to analyze the effects of the quadruple expression cassettes (Fig. 5a) *in planta*, we successfully demonstrated that multiple transgenic rice lines with these cassettes exhibit strong tolerance to low Fe availability in calcareous soils (Fig. 8; Fig. S8, Fig. S9). Notably, this tolerance was comparable to (line Q1) or even higher than (line Q7) that of previously selected highly tolerant lines with double (line D) or triple (line T) expression cassettes (Fig. 8; Masuda et al. 2017, 2019). Further selection of tolerant lines among the quadruple lines would lead to lines with still higher tolerance of low Fe availability being obtained, because we used only 12 lines for the analysis of Q lines (Fig. S8), whereas lines D and T used for our controls (Fig. 8) have been selected as the best performing ones among 28 and 73 regenerated lines, respectively (Masuda et al. 2017, 2019). Phenotypes caused by low Fe availability are affected by various environmental factors (Suzuki et al., 2008; Masuda et al. 2019). Our recent analyses in transgenic rice revealed that reinforcement of the strategy II system is highly effective under aerobic conditions without submergence, whereas this effect is severely impaired under submerged conditions (Masuda et al. 2019; Kobayashi et al. 2022a). This effect is thought to be due to dilution of increased MAs from the rhizosphere. Meanwhile, the function of Fe deficiency-inducible *Refre1/372* introduced into rice is more prominent in submerged conditions (Masuda et al. 2019; Kobayashi et al. 2022a), in which reduced Fe^{2+} could be effectively maintained for subsequent uptake by OsIRT1 transporter. Although calcareous soil is mainly distributed in arid and semiarid regions, simultaneous enhancement of strategy II and partial strategy I by introducing multiple genes would be favorable in fluctuating water conditions. Such

simultaneous enhancement is achieved by our double, triple, and quadruple expression cassettes (Fig. 5a; Masuda et al. 2017, 2019). Recently, we also showed that the enhancement of endogenous Fe uptake and translocation in rice by knocking down *OsHRZ* negative regulators with the concomitant introduction of an exogenous Fe uptake system by introduction of the *OsIRT1* promoter-*Refre1/372* cassette is highly effective for conferring tolerance of Fe deficiency in submerged and nonsubmerged conditions with the high accumulation of Fe and Zn in seeds (Kobayashi et al. 2022a). This could in turn help to prevent human Fe and Zn deficiencies. In addition, higher production of NA is suggested to confer drought tolerance in rice (Lee et al. 2017). More generally, pyramiding of multiple traits would increase the robustness of plant stress tolerance, resulting in the possibility of better performance in variable environmental conditions. In summary, our results provide a basis for the future bioengineering of sorghum and other plants with increased bioproduction in Fe-limited environments.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11103-024-01508-y>.

Acknowledgements We thank Ms. Hiroko Hori, Ms. Mayumi Ura, and Ms. Kazuko Arai (Ishikawa Prefectural University) for their assistance with plant culture and analysis.

Author contributions T.S. carried out most of the research on wild-type sorghum, constructed the quadruple vector, produced, and analyzed the Q rice lines, and wrote the draft of the manuscript. T.N. analyzed the enzymatic activities of SbNAS1 and MA secretion of sorghum, and carried out intercropping experiments. R.Y. produced the quadruple vector and the Q rice lines, and analyzed them. M.Y. and H.N. analyzed the transgenic sorghum lines. K.M. carried out the calcareous soil culture of rice Q lines in comparison with D and T lines. K.I. and H.E. produced transgenic sorghum lines and carried out Southern hybridization analysis. R.N.I. supported the analysis of MA secretion. K.B. assisted the production of recombinant SbNAS1. H.M. supported the analysis of calcareous soil culture. T.K. carried out metal concentration analysis in rice, supervised the rice research, analyzed data, and wrote, and revised the manuscript with discussion from all of the co-authors. H.N. and N.K.N. designed and supervised the research project.

Funding This research was supported by the Advanced Low Carbon Technology Research and Development Program (ALCA) from the Japan Science and Technology Agency to N.K.N., H.N., and T.K. This research was also partly supported by a grant from Urugami-zaidan to T.N. and R.N.I., and by the Japan Society for the Promotion of Sciences (JSPS) KAKENHI, with Grant Numbers JP15K18658, JP15KK0286, and JP20K05777 to T.N., and Grant Numbers JP18H02115 and JP22H02231 to T.K.

Data availability Enquiries about data availability should be directed to the authors.

Declarations

Conflict of interests The authors declare no competing interests.

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